Tables, keys and types

Data engineering · Level 3 · Lesson 1



In this lesson you will

- Explain why apps use a database.
- Design tables with primary keys, types and foreign keys.
- Write a CREATE TABLE and a first SELECT.

Key ideas

- A primary key picks out one row.
- Foreign keys link tables.
- A good schema stores each fact once.

Split the wide table

1. Look (3 min): one wide table
2. Split (6 min): two tables, one key
3. Change (4 min): a new phone number
4. Write (4 min): CREATE TABLE

Use made-up names and numbers.

Let’s discuss

1. Why are names bad primary keys?
2. What went wrong when a fact was stored twice?
3. Which column in likes points to photos?

Exit ticket

- Sketch two linked tables for an app, and mark the primary and foreign keys.